DRC Ebola Outbreak Prediction — Weekly Executive Report

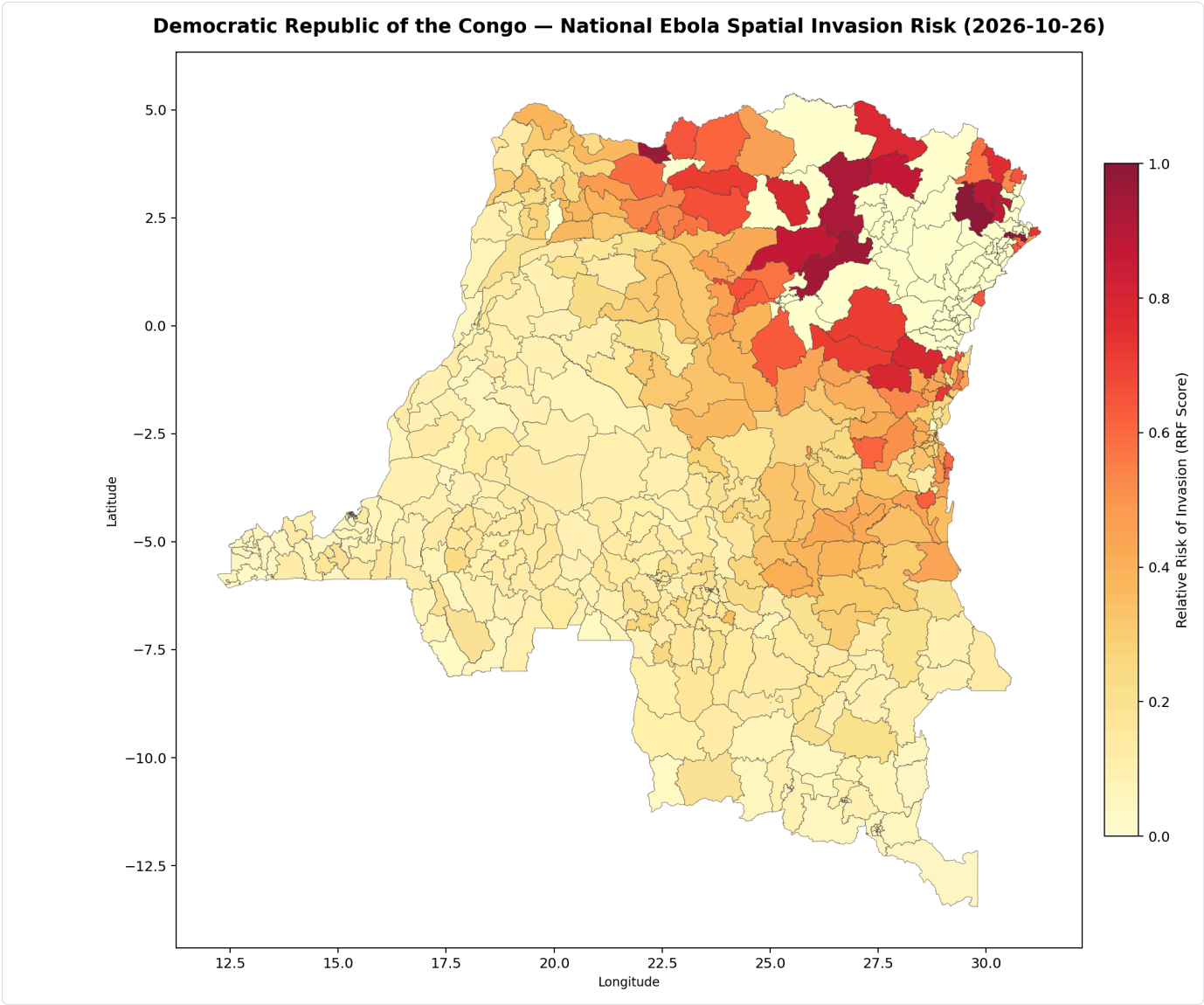
1. National Forecast & Top 25 Surveillance Tiers

Forecast Window: `2026-10-27` to `2026-11-02` | **Cutoff Date**: `2026-10-26`

Data Freshness: sitrep cases through `2026-10-21` (5 days before cutoff) | INRB estimate cutoff `2026-10-26` (0 days before cutoff)

Primary Ranking: `Google AI + INRB Reciprocal Rank Fusion (RRF k=60)` | **Ensemble**: 20 bootstrap seeds (95% empirical CI)

Optimal Config: `leaves=31`, `l2=0.0`, `lr=0.005`, `leaf=20`, `iter=300`



Top 25 Health Zones Ranked by RRF Multi-Model Relative Risk

RRF Rank	Google Rank	INRB Rank	Health Zone	Province	Ensemble Score	INRB Pred	95% CI	Vulnerability (V)	Priority Rank	Top Contributing Features
1	7	2	**Watsa**	Haut-Uele	0.0118	0.0215	[0.0027 - 0.0287]	0.640	9	Flowminder Telecom Flow Pressur (+26.3%), Travel Time to Active Infe (+10.9%), Long-Range Intercity Gra (-8.5%), Source Epicenter Case Gro (-7.8%), Local Economic Activity (G (-7.1%)
2	9	1	**Rethy**	Ituri	0.0105	0.0386	[0.0027 - 0.0228]	0.360	14	Intra-Provincial Transmission Press (-26.3%), Travel Time to Active Infe (+19.2%), Spatial Gravity Hazard (+ Flowminder Telecom Flow Pressur (+6.6%), Mining Corridor Flow Haza (+5.3%)
3	3	11	**Yakoma**	Nord-Ubangi	0.0275	0.0082	[0.0028 - 0.0552]	0.885	3	Flowminder Telecom Flow Pressur (+19.5%), Travel Time to Active Infe (+19.2%), Transit Hub Connectivity Gravity Hazard (Power Law) (+5.8% Socioeconomic Vulnerability Index
4	6	9	**Wasolo**	Nord-Ubangi	0.0134	0.0133	[0.0027 - 0.0257]	0.761	6	Travel Time to Active Infection (+2 Long-Range Intercity Gravity (+14.6 Socioeconomic Vulnerability Index Flowminder Telecom Flow Pressur (-6.8%), Gravity Hazard (Power Law (+6.7%)
5	1	16	**Bafwagbogbo**	Tshopo	0.1145	0.0049	[0.0039 - 0.2170]	0.526	1	Flowminder Telecom Flow Pressur (+32.0%), Population Dynamics Em (PDFM) (-10.5%), Travel Time to Ac Infection (+9.6%), Spatial Gravity Ha (+4.5%), Transit Hub Connectivity (-
6	10	8	**Poko**	Bas-Uele	0.0104	0.0143	[0.0027 - 0.0198]	0.969	8	Travel Time to Active Infection (+21 Flowminder Telecom Flow Pressur (+16.9%), Long-Range Intercity Gra (-15.0%), Spatial Gravity Hazard (+6 Socioeconomic Vulnerability Index
7	11	12	**Makoro**	Haut-Uele	0.0102	0.0081	[0.0027 - 0.0214]	0.510	11	Travel Time to Active Infection (+1 Long-Range Intercity Gravity (-11.3 Economic Activity (GDP pc) (-10.0% Flowminder Telecom Flow Pressur (+9.4%), Spatial Gravity Hazard (+6
8	20	6	**Biringi**	Ituri	0.0066	0.0149	[0.0027 - 0.0146]	0.466	20	Intra-Provincial Transmission Press (-25.8%), Travel Time to Active Infe (+14.9%), Flowminder Telecom Flo Pressure (+9.3%), Long-Range Inter Gravity (-8.0%), Spatial Gravity Haza (+4.9%)
9	13	13	**Banalia**	Tshopo	0.0092	0.0073	[0.0027 - 0.0199]	0.566	13	Travel Time to Active Infection (+17 Flowminder Telecom Flow Pressur (+15.7%), Population Dynamics Em (PDFM) (-15.1%), Long-Range Inter Gravity (-11.2%), Population Mobilit Pressure (+4.8%)
10	27	4	**Niangara**	Haut-Uele	0.0053	0.0158	[0.0027 - 0.0088]	0.735	23	Travel Time to Active Infection (+20 Flowminder Telecom Flow Pressur (+17.0%), Long-Range Intercity Gra (-14.6%), Spatial Gravity Hazard (+5 Population Mobility Pressure (+5.1
11	4	44	**Titule**	Bas-Uele	0.0266	0.0015	[0.0040 - 0.0587]	0.912	4	Travel Time to Active Infection (+18 Long-Range Intercity Gravity (+15.5 Population Dynamics Embedding (I (+14.8%), Local Economic Activity ((-7.1%), Secondary Diffusion Pressu (+6.2%)
12	5	43	**Pinga**	Nord-Kivu	0.0186	0.0015	[0.0031 - 0.0353]	0.760	5	Travel Time to Active Infection (+16 Flowminder Telecom Flow Pressur (+12.7%), Population Dynamics Em (PDFM) (-9.9%), Long-Range Interci Gravity (-7.4%), Spatial Gravity Haza (+7.3%)
13	16	24	**Walikale**	Nord-Kivu	0.0086	0.0027	[0.0027 -	0.764	12	Flowminder Telecom Flow Pressur (+17.0%), Travel Time to Active Infe

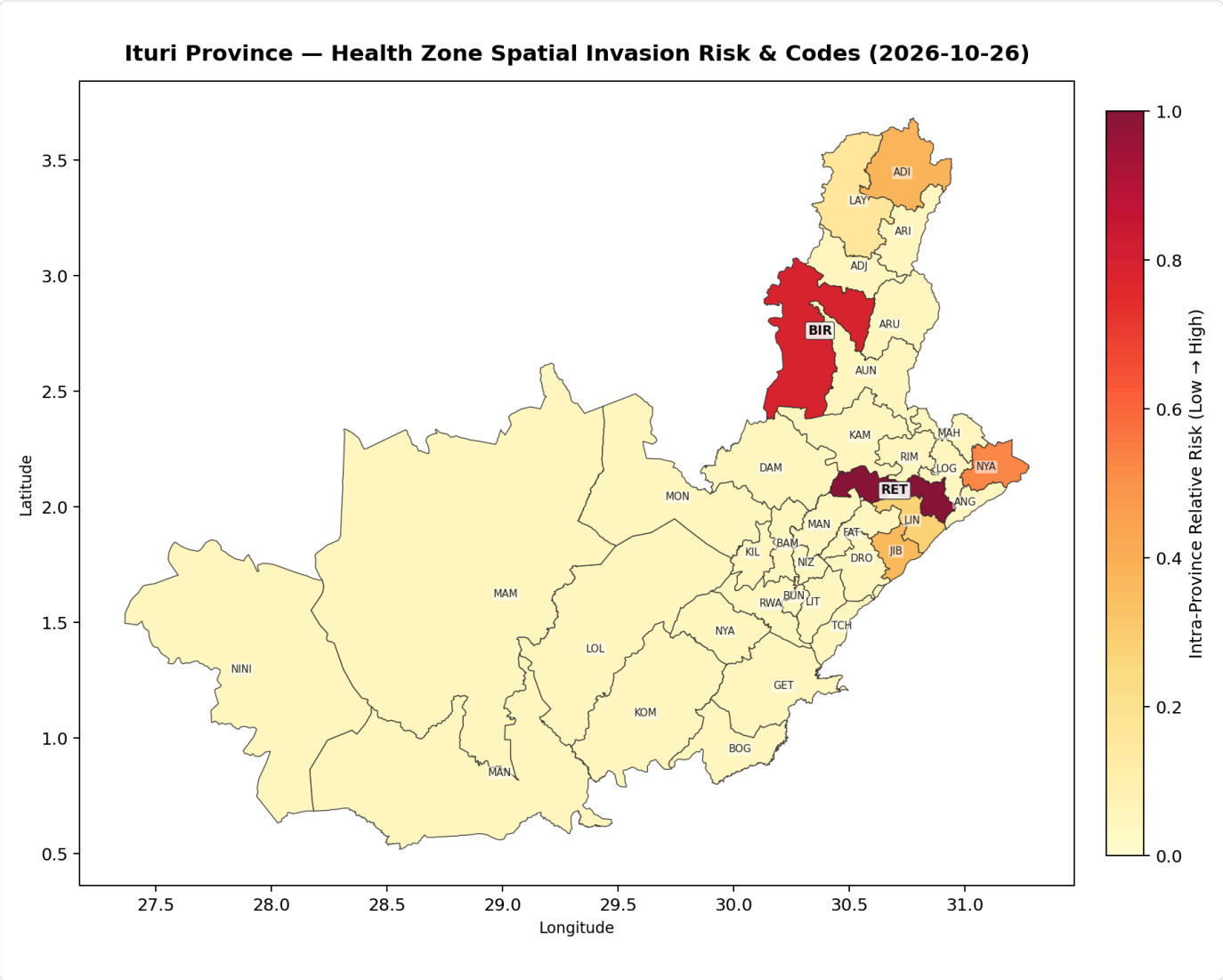
RRF Rank	Google Rank	INRB Rank	Health Zone	Province	Ensemble Score	INRB Pred	95% CI	Vulnerability (V)	Priority Rank	Top Contributing Features
							0.0164]			(+16.6%), Population Dynamics Em (PDFM) (-12.8%), Long-Range Inter Gravity (-8.1%), Spatial Gravity Haza (+7.2%)
14	8	37	**Doruma**	Haut-Uele	0.0113	0.0018	[0.0029 - 0.0215]	0.829	7	Travel Time to Active Infection (+19 Secondary Diffusion Pressure (+17 Long-Range Intercity Gravity (+11.0 Local Economic Activity (GDP pc) (Flowminder Telecom Flow Pressur
15	30	15	**Aba**	Haut-Uele	0.0052	0.0060	[0.0027 - 0.0107]	0.759	26	Travel Time to Active Infection (+22 Local Economic Activity (GDP pc) (Long-Range Intercity Gravity (-11.0 Spatial Gravity Hazard (+10.0%), Sc Epicenter Case Growth (+5.4%)
16	22	27	**Nyarambe**	Ituri	0.0062	0.0023	[0.0027 - 0.0127]	0.241	33	Intra-Provincial Transmission Press (-18.8%), Travel Time to Active Infe (+17.1%), Socioeconomic Vulnerab Index (-15.9%), Flowminder Telecor Pressure (+8.0%), Spatial Gravity Ha (+3.8%)
17	17	35	**Kirotshe**	Nord-Kivu	0.0086	0.0018	[0.0027 - 0.0188]	0.512	15	Travel Time to Active Infection (+24 Long-Range Intercity Gravity (-12.9 Flowminder Telecom Flow Pressur (-10.4%), Distance to Nearest Activ (+9.0%), Socioeconomic Vulnerabil (-4.0%)
18	25	28	**Lubutu**	Maniema	0.0055	0.0022	[0.0027 - 0.0100]	0.627	27	Travel Time to Active Infection (+19 Secondary Diffusion Pressure (-10. Spatial Gravity Hazard (+7.7%), Flo Telecom Flow Pressure (+6.8%), Ur Settlement Fraction (-6.4%)
19	2	83	**Likati**	Bas-Uele	0.0340	0.0006	[0.0027 - 0.0749]	0.950	2	Travel Time to Active Infection (+21 Flowminder Telecom Flow Pressur (+15.5%), Long-Range Intercity Grav (+11.8%), Population Dynamics Em (PDFM) (+8.0%), Transit Hub Conne (+7.5%)
20	24	30	**Opienge**	Tshopo	0.0055	0.0020	[0.0027 - 0.0115]	0.593	28	Travel Time to Active Infection (+20 Population Dynamics Embedding (I (-14.0%), Long-Range Intercity Grav (+8.7%), Flowminder Telecom Flow Pressure (-7.6%), Spatial Gravity Ha (+6.7%)
21	19	47	**Kadutu**	Sud-Kivu	0.0068	0.0014	[0.0027 - 0.0149]	0.218	30	Travel Time to Active Infection (+20 Flowminder Telecom Flow Pressur (-12.3%), Distance to Nearest Activ (+8.4%), Secondary Diffusion Press (-6.9%), Long-Range Intercity Gravit
22	50	19	**Isangi**	Tshopo	0.0037	0.0038	[0.0027 - 0.0082]	0.512	52	Travel Time to Active Infection (+20 Population Dynamics Embedding (I (-17.9%), Long-Range Intercity Grav (-10.5%), Distance to Outbreak Epic (-6.3%), Transit Hub Connectivity (+
23	15	61	**Aketi**	Bas-Uele	0.0089	0.0008	[0.0027 - 0.0187]	0.902	10	Travel Time to Active Infection (+29 Long-Range Intercity Gravity (+16.4 Flowminder Telecom Flow Pressur (-7.8%), Secondary Diffusion Pressu (+7.3%), Population Dynamics Emb (PDFM) (-6.0%)
24	82	10	**Adi**	Ituri	0.0027	0.0100	[0.0027 - 0.0046]	0.530	247	Intra-Provincial Transmission Press (-28.1%), Travel Time to Active Infe (+19.7%), Long-Range Intercity Grav (-10.4%), Spatial Gravity Hazard (+5 Population Dynamics Embedding (I (+3.9%)
25	35	34	**Monga**	Bas-Uele	0.0044	0.0019	[0.0027 - 0.0091]	0.978	29	Travel Time to Active Infection (+23 Long-Range Intercity Gravity (+12.8 Flowminder Telecom Flow Pressur (-7.9%), Population Mobility Pressu (-7.6%), Socioeconomic Vulnerabili (+7.6%)

** Top Contributing Features show signed percentage shares relative to total absolute feature attribution (100% × c_j / ∑|c_k|), reflecting the direction (+/-) and relative magnitude of each driver on model risk.*

2. Priority Province Spatial Risk Heatmaps & Top 10 Health Zones

Province-specific spatial invasion risk heatmaps and top 10 health zones, ranked by multi-model Reciprocal Rank Fusion (RRF).

Ituri Province

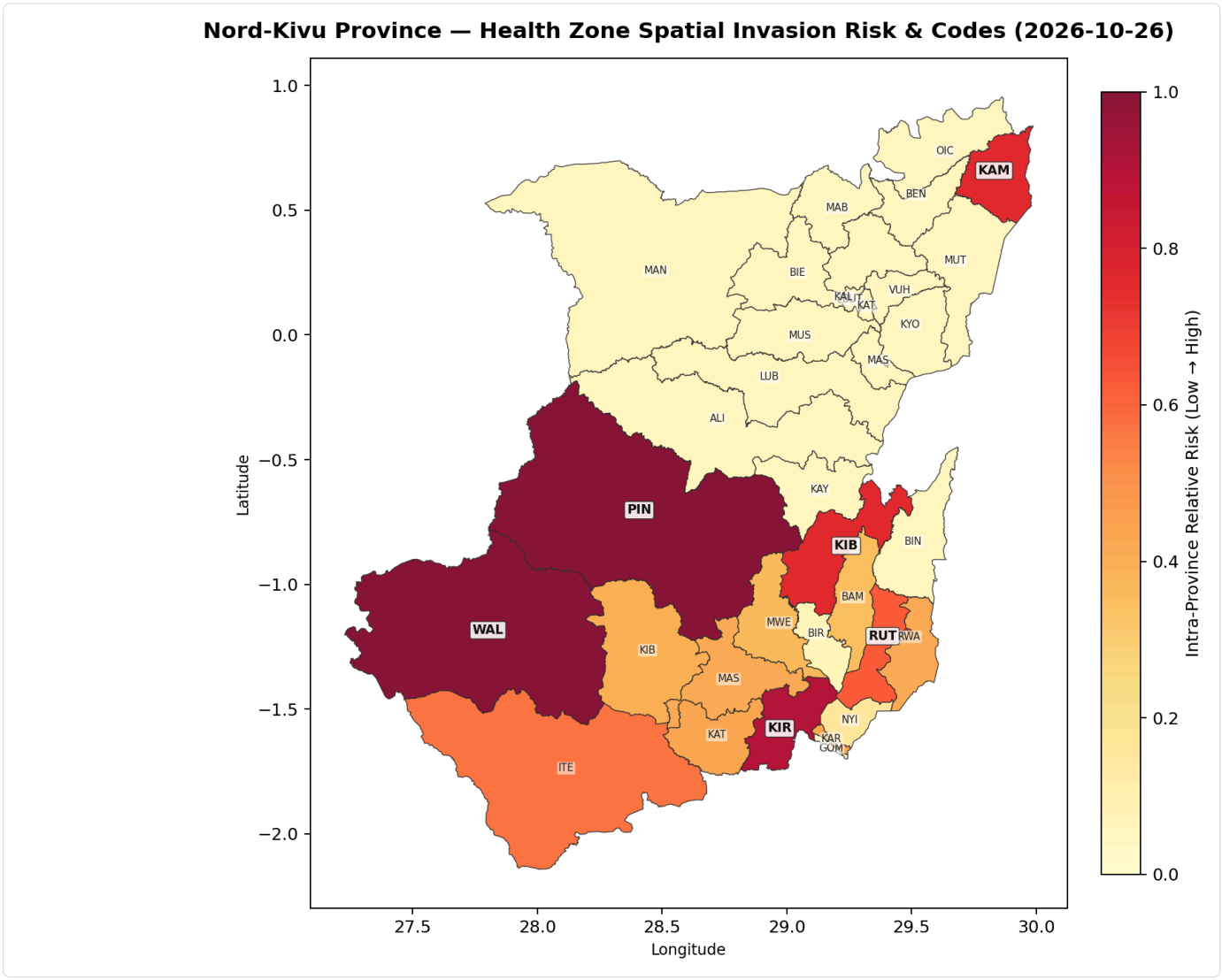


Top 10 Health Zones in Ituri (Ranked by RRF)

Prov #	RRF Rank	Google Rank	INRB Rank	Health Zone	Ensemble Score	INRB Pred	95% CI	Vulnerability (V)	Priority Rank	Top Contributing Features
1	**2**	9	1	**Rethy**	0.0105	0.0386	[0.0027 - 0.0228]	0.360	14	Intra-Provincial Transmission Pressure (-26.3%), Travel Time to Active Infection (+19.2%), Spatial Gravity Hazard (+7.9% Flowminder Telecom Flow Pressure (+6.6%), Mining Corridor Flow Hazard (+5.3%)
2	**8**	20	6	**Biringi**	0.0066	0.0149	[0.0027 - 0.0146]	0.466	20	Intra-Provincial Transmission Pressure (-25.8%), Travel Time to Active Infection (+14.9%), Flowminder Telecom Flow Pressure (+9.3%), Long-Range Intercity Gravity (-8.0%), Spatial Gravity Hazard (+4.9%)
3	**16**	22	27	**Nyarambe**	0.0062	0.0023	[0.0027 - 0.0127]	0.241	33	Intra-Provincial Transmission Pressure (-18.8%), Travel Time to Active Infection (+17.1%), Socioeconomic Vulnerability Index (-15.9%), Flowminder Telecom Flc Pressure (+8.0%), Spatial Gravity Hazar (+3.8%)
4	**24**	82	10	**Adi**	0.0027	0.0100	[0.0027 - 0.0046]	0.530	247	Intra-Provincial Transmission Pressure (-28.1%), Travel Time to Active Infection (+19.7%), Long-Range Intercity Gravity (-10.4%), Spatial Gravity Hazard (+5.0%) Population Dynamics Embedding (PDFM (+3.9%)
5	**26**	129	3	**Jiba**	0.0027	0.0178	[0.0027 - 0.0032]	0.220	403	Intra-Provincial Transmission Pressure (-26.8%), Travel Time to Active Infection (+19.8%), Spatial Gravity Hazard (+7.5% Population Mobility Pressure (-5.2%), Flowminder Telecom Flow Pressure (-5.
6	**37**	139	7	**Linga**	0.0027	0.0149	[0.0027 - 0.0027]	0.286	394	Intra-Provincial Transmission Pressure (-27.3%), Travel Time to Active Infection (+19.7%), Spatial Gravity Hazard (+7.7% Flowminder Telecom Flow Pressure (-6.1%), Major Highway Corridor Gravity (+4.1%)
7	**43**	159	14	**Laybo**	0.0027	0.0071	[0.0027 - 0.0029]	0.452	305	Intra-Provincial Transmission Pressure (-26.1%), Travel Time to Active Infection (+19.1%), Long-Range Intercity Gravity (-8.1%), Population Mobility Pressure (-5.5%), Flowminder Telecom Flow Pressure (+5.3%)
8	**51**	81	48	**Angumu**	0.0027	0.0013	[0.0027 - 0.0049]	0.147	416	Intra-Provincial Transmission Pressure (-18.8%), Travel Time to Active Infection (+16.6%), Socioeconomic Vulnerability Index (-15.2%), Flowminder Telecom Flc Pressure (+7.7%), Long-Range Intercity Gravity (-5.5%)

** Top Contributing Features show signed percentage shares relative to total absolute feature attribution (100% × c_j / ∑|c_k|).*

Nord-Kivu Province

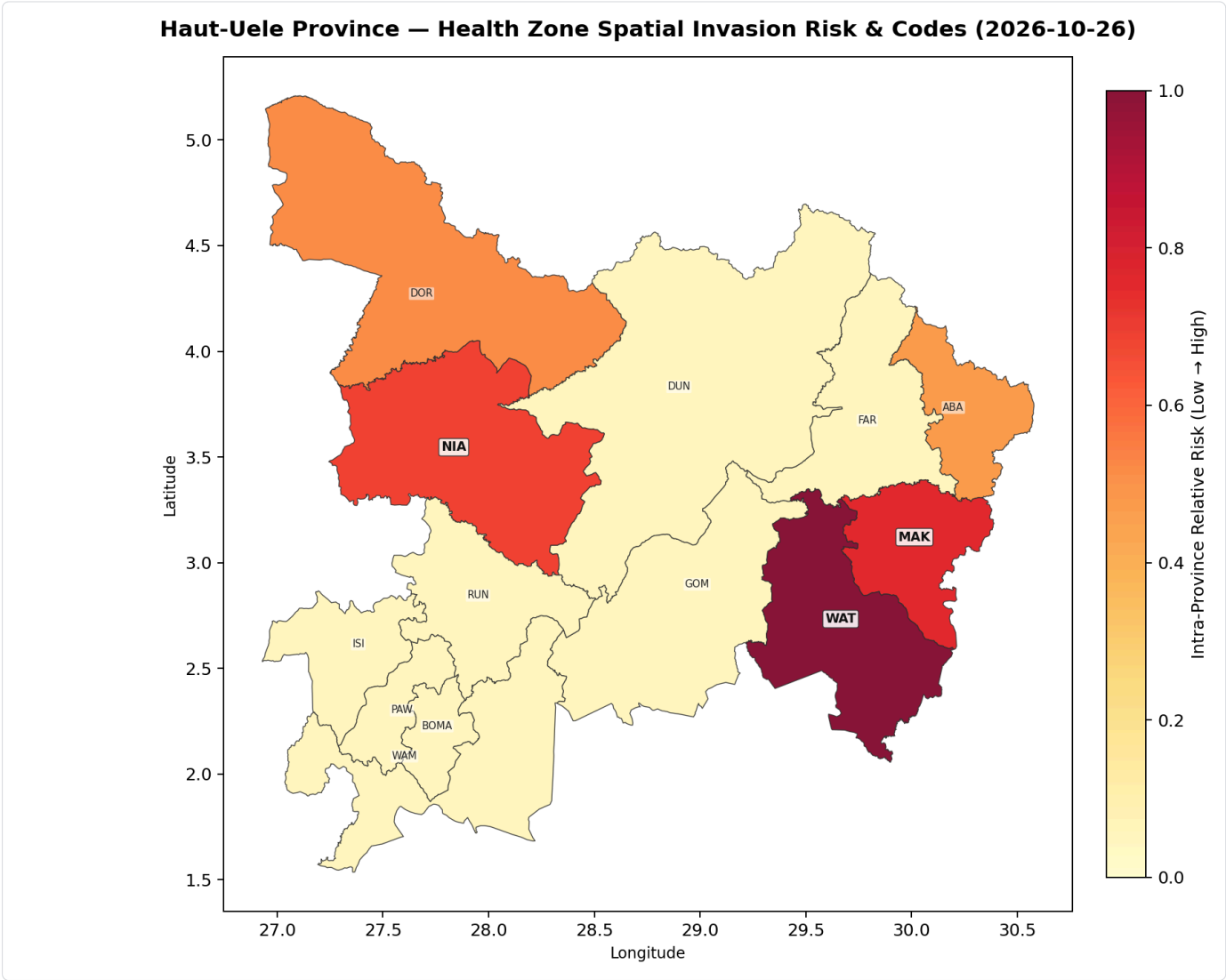


Top 10 Health Zones in Nord-Kivu (Ranked by RRF)

Prov #	RRF Rank	Google Rank	INRB Rank	Health Zone	Ensemble Score	INRB Pred	95% CI	Vulnerability (V)	Priority Rank	Top Contributing Features
1	**12**	5	43	**Pinga**	0.0186	0.0015	[0.0031 - 0.0353]	0.760	5	Travel Time to Active Infection (+16.4%), Flowminder Telecom Flow Pressure (+12.7%), Population Dynamics Embedd (PDFM) (-9.9%), Long-Range Intercity Gravity (-7.4%), Spatial Gravity Hazard (+7.3%)
2	**13**	16	24	**Walikale**	0.0086	0.0027	[0.0027 - 0.0164]	0.764	12	Flowminder Telecom Flow Pressure (+17.0%), Travel Time to Active Infectio (+16.6%), Population Dynamics Embedd (PDFM) (-12.8%), Long-Range Intercity Gravity (-8.1%), Spatial Gravity Hazard (+7.2%)
3	**17**	17	35	**Kirotshe**	0.0086	0.0018	[0.0027 - 0.0188]	0.512	15	Travel Time to Active Infection (+24.6%), Long-Range Intercity Gravity (-12.9%), Flowminder Telecom Flow Pressure (-10.4%), Distance to Nearest Active Zo (+9.0%), Socioeconomic Vulnerability In (-4.0%)
4	**27**	60	18	**Kibirizi**	0.0032	0.0042	[0.0027 - 0.0063]	0.557	73	Travel Time to Active Infection (+22.4%), Long-Range Intercity Gravity (-12.2%), Distance to Outbreak Epicenter (+9.8%), Flowminder Telecom Flow Pressure (-8.8%), Population Mobility Pressure (-5.6%)
5	**28**	51	23	**Kamango**	0.0037	0.0027	[0.0027 - 0.0065]	0.583	49	Travel Time to Active Infection (+14.3%), Spatial Gravity Hazard (+11.4%), Major Highway Corridor Gravity (+8.1%), Dista to Outbreak Epicenter (+7.7%), Populati Dynamics Embedding (PDFM) (-7.0%)
6	**42**	49	42	**Rutshuru**	0.0037	0.0015	[0.0027 - 0.0086]	0.496	53	Travel Time to Active Infection (+21.1%), Socioeconomic Vulnerability Index (-20.9%), Distance to Outbreak Epicente (+8.6%), Flowminder Telecom Flow Pressure (-6.8%), Source Epicenter Case Growth (-4.0%)
7	**44**	28	90	**Itebero**	0.0053	0.0006	[0.0027 - 0.0121]	0.743	24	Travel Time to Active Infection (+24.4%), Flowminder Telecom Flow Pressure (-9.3%), Population Mobility Pressure (-8.7%), Long-Range Intercity Gravity (+5.9%), Urban Settlement Fraction (-5.8
8	**56**	110	41	**Katoyi**	0.0027	0.0016	[0.0027 - 0.0038]	0.606	176	Travel Time to Active Infection (+21.1%), Long-Range Intercity Gravity (-13.6%), Flowminder Telecom Flow Pressure (-8.8%), Population Mobility Pressure (-7.3%), Spatial Gravity Hazard (+6.2%)
9	**59**	58	79	**Karisimbi**	0.0034	0.0007	[0.0027 - 0.0076]	0.213	253	Travel Time to Active Infection (+18.3%), Socioeconomic Vulnerability Index (-17.6%), Long-Range Intercity Gravity (-10.3%), Distance to Nearest Active Zo (+6.6%), Google Aggregated Flow Press (+6.1%)
10	**63**	194	26	**Masisi**	0.0027	0.0024	[0.0027 - 0.0027]	0.587	191	Travel Time to Active Infection (+25.4%), Long-Range Intercity Gravity (-12.6%), Secondary Diffusion Pressure (-10.4%), Flowminder Telecom Flow Pressure (-9.3%), Spatial Gravity Hazard (+4.4%)

** Top Contributing Features show signed percentage shares relative to total absolute feature attribution (100% × c_j / ∑|c_k|).*

Haut-Uele Province

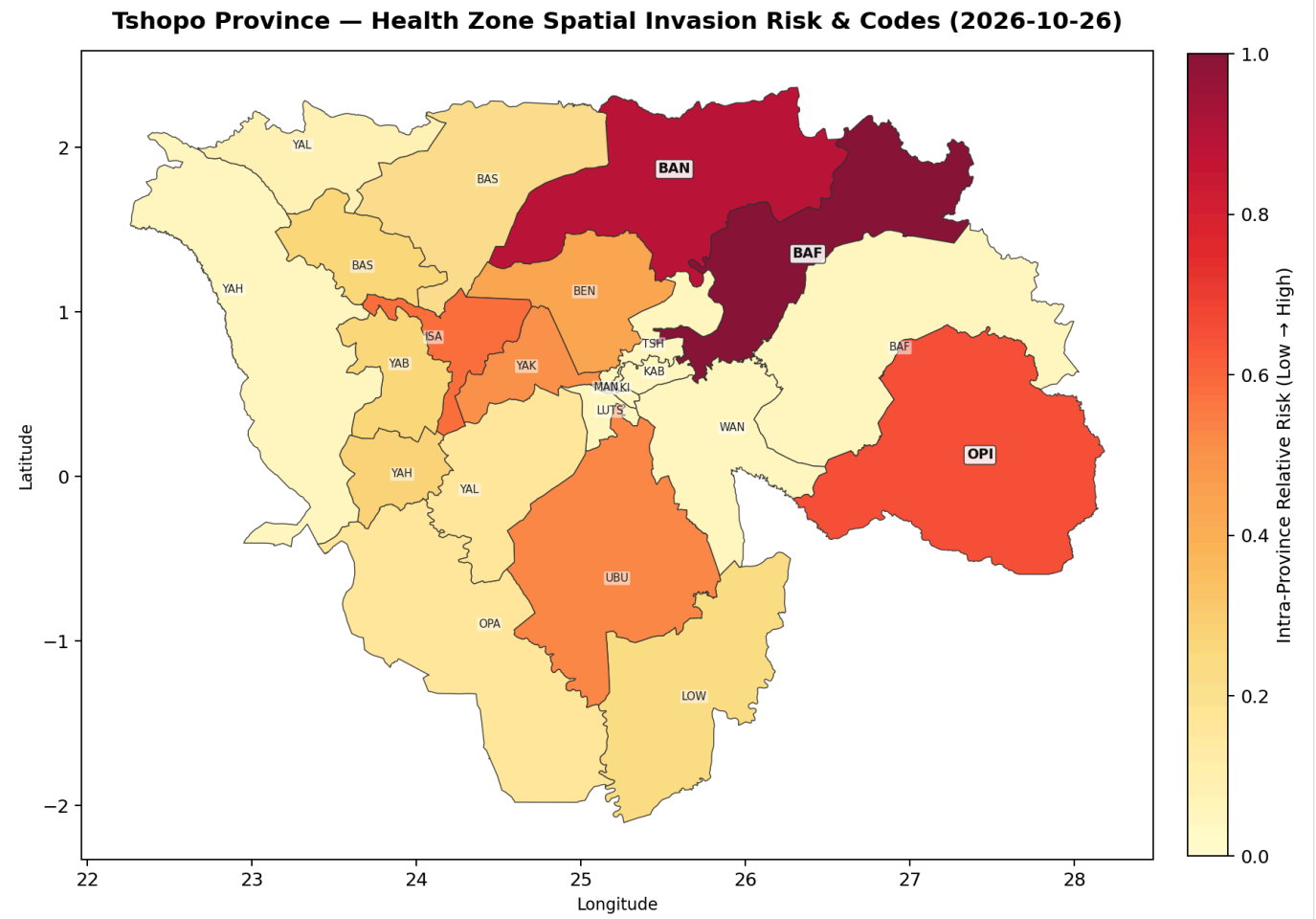


Top 10 Health Zones in Haut-Uele (Ranked by RRF)

Prov #	RRF Rank	Google Rank	INRB Rank	Health Zone	Ensemble Score	INRB Pred	95% CI	Vulnerability (V)	Priority Rank	Top Contributing Features
1	**1**	7	2	**Watsa**	0.0118	0.0215	[0.0027 - 0.0287]	0.640	9	Flowminder Telecom Flow Pressure (+26.3%), Travel Time to Active Infectio (+10.9%), Long-Range Intercity Gravity (-8.5%), Source Epicenter Case Growth (-7.8%), Local Economic Activity (GDP p (-7.1%)
2	**7**	11	12	**Makoro**	0.0102	0.0081	[0.0027 - 0.0214]	0.510	11	Travel Time to Active Infection (+15.9%) Long-Range Intercity Gravity (-11.3%), L Economic Activity (GDP pc) (-10.0%), Flowminder Telecom Flow Pressure (+9.4%), Spatial Gravity Hazard (+6.9%)
3	**10**	27	4	**Niangara**	0.0053	0.0158	[0.0027 - 0.0088]	0.735	23	Travel Time to Active Infection (+20.5%) Flowminder Telecom Flow Pressure (+17.0%), Long-Range Intercity Gravity (-14.6%), Spatial Gravity Hazard (+5.7%) Population Mobility Pressure (+5.1%)
4	**14**	8	37	**Doruma**	0.0113	0.0018	[0.0029 - 0.0215]	0.829	7	Travel Time to Active Infection (+19.4%) Secondary Diffusion Pressure (+17.9%), Long-Range Intercity Gravity (+11.0%), Local Economic Activity (GDP pc) (-8.3 Flowminder Telecom Flow Pressure (-7.
5	**15**	30	15	**Aba**	0.0052	0.0060	[0.0027 - 0.0107]	0.759	26	Travel Time to Active Infection (+22.0%) Local Economic Activity (GDP pc) (-11.1 Long-Range Intercity Gravity (-11.0%), Spatial Gravity Hazard (+10.0%), Source Epicenter Case Growth (+5.4%)
6	**41**	75	25	**Faradje**	0.0027	0.0027	[0.0027 - 0.0036]	0.597	187	Travel Time to Active Infection (+18.6%) Local Economic Activity (GDP pc) (-11.8 Population Dynamics Embedding (PDF (-8.4%), Spatial Gravity Hazard (+8.1%), Long-Range Intercity Gravity (-5.2%)

** Top Contributing Features show signed percentage shares relative to total absolute feature attribution (100% × c_j / ∑|c_k|).*

Tshopo Province

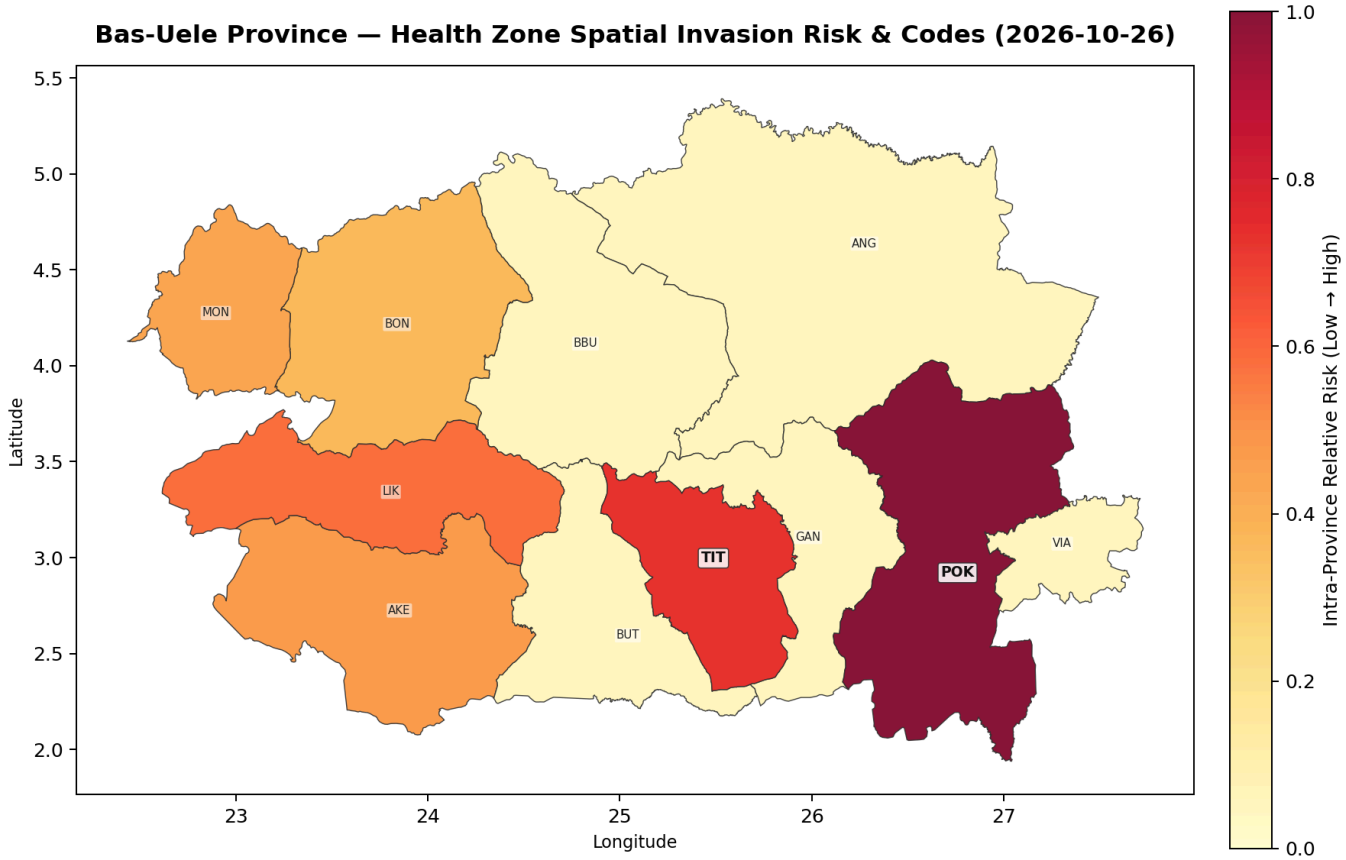


Top 10 Health Zones in Tshopo (Ranked by RRF)

Prov #	RRF Rank	Google Rank	INRB Rank	Health Zone	Ensemble Score	INRB Pred	95% CI	Vulnerability (V)	Priority Rank	Top Contributing Features
1	**5**	1	16	**Bafwagbogbo**	0.1145	0.0049	[0.0039 - 0.2170]	0.526	1	Flowminder Telecom Flow Pressure (+32.0%), Population Dynamics Embe (PDFM) (-10.5%), Travel Time to Active Infection (+9.6%), Spatial Gravity Haza (+4.5%), Transit Hub Connectivity (+4.
2	**9**	13	13	**Banalia**	0.0092	0.0073	[0.0027 - 0.0199]	0.566	13	Travel Time to Active Infection (+17.3 Flowminder Telecom Flow Pressure (+15.7%), Population Dynamics Embe (PDFM) (-15.1%), Long-Range Intercity Gravity (-11.2%), Population Mobility Pressure (+4.8%)
3	**20**	24	30	**Opienge**	0.0055	0.0020	[0.0027 - 0.0115]	0.593	28	Travel Time to Active Infection (+20.9 Population Dynamics Embedding (PDI (-14.0%), Long-Range Intercity Gravity (+8.7%), Flowminder Telecom Flow Pressure (-7.6%), Spatial Gravity Haza (+6.7%)
4	**22**	50	19	**Isangi**	0.0037	0.0038	[0.0027 - 0.0082]	0.512	52	Travel Time to Active Infection (+20.7 Population Dynamics Embedding (PDI (-17.9%), Long-Range Intercity Gravity (-10.5%), Distance to Outbreak Epicen (-6.3%), Transit Hub Connectivity (+6.1
5	**29**	41	32	**Ubundu**	0.0040	0.0020	[0.0027 - 0.0078]	0.633	40	Travel Time to Active Infection (+23.5 Population Dynamics Embedding (PDI (-13.9%), Distance to Outbreak Epicen (-8.0%), Long-Range Intercity Gravity (-6.4%), Transit Hub Connectivity (+5.3
6	**31**	138	5	**Yakusu**	0.0027	0.0154	[0.0027 - 0.0029]	0.425	331	Travel Time to Active Infection (+28.2 Population Dynamics Embedding (PDI (-13.7%), Flowminder Telecom Flow Pressure (-7.7%), Distance to Outbreak Epicenter (-5.6%), Long-Range Intercity Gravity (-5.5%)
7	**39**	86	20	**Bengamisa**	0.0027	0.0033	[0.0027 - 0.0037]	0.402	347	Travel Time to Active Infection (+24.6 Secondary Diffusion Pressure (-9.3%), Distance to Outbreak Epicenter (-6.4% Transit Hub Connectivity (+5.2%), Lon Range Intercity Gravity (-5.2%)
8	**53**	79	53	**Yahisuli**	0.0027	0.0010	[0.0027 - 0.0040]	0.469	295	Travel Time to Active Infection (+26.5 Secondary Diffusion Pressure (-7.3%), Socioeconomic Vulnerability Index (+7 Population Mobility Pressure (-5.4%), International Border Proximity (-5.1%)
9	**54**	74	57	**Basoko**	0.0027	0.0009	[0.0027 - 0.0039]	0.402	347	Travel Time to Active Infection (+27.0 Long-Range Intercity Gravity (-7.4%), Population Mobility Pressure (+7.3%), Mining Corridor Flow Hazard (-5.6%), International Border Proximity (-5.1%)
10	**55**	151	29	**Yabaondo**	0.0027	0.0020	[0.0027 - 0.0027]	0.470	293	Travel Time to Active Infection (+29.7 Secondary Diffusion Pressure (-6.1%), Distance to Outbreak Epicenter (-5.8% Mining Corridor Flow Hazard (-5.6%), L Range Intercity Gravity (-5.3%)

** Top Contributing Features show signed percentage shares relative to total absolute feature attribution (100% × c_j / ∑|c_k|).*

Bas-Uele Province

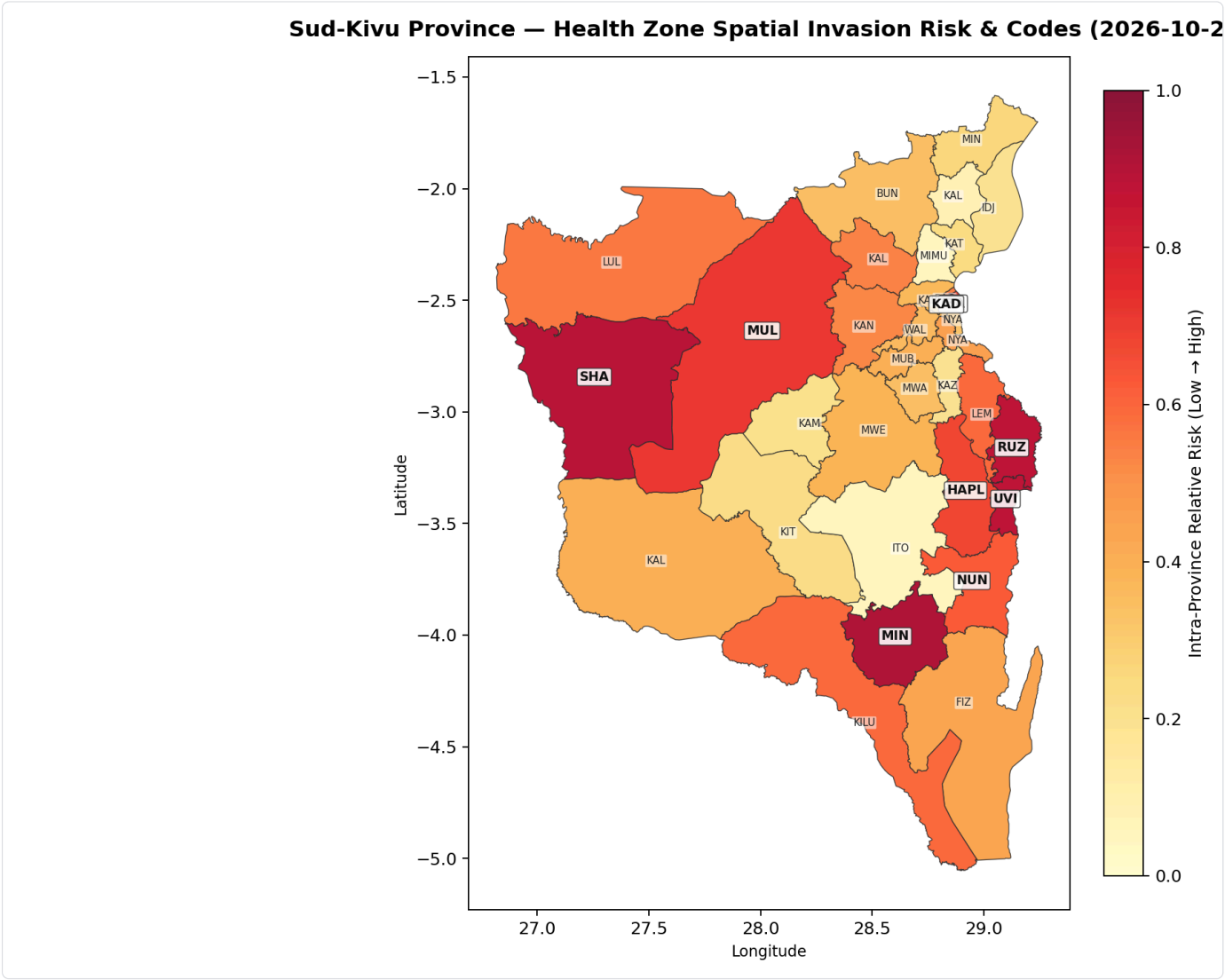


Top 10 Health Zones in Bas-Uele (Ranked by RRF)

Prov #	RRF Rank	Google Rank	INRB Rank	Health Zone	Ensemble Score	INRB Pred	95% CI	Vulnerability (V)	Priority Rank	Top Contributing Features
1	**6**	10	8	**Poko**	0.0104	0.0143	[0.0027 - 0.0198]	0.969	8	Travel Time to Active Infection (+21.5%), Flowminder Telecom Flow Pressure (+16.9%), Long-Range Intercity Gravity (-15.0%), Spatial Gravity Hazard (+6.5%), Socioeconomic Vulnerability Index (+5.7
2	**11**	4	44	**Titule**	0.0266	0.0015	[0.0040 - 0.0587]	0.912	4	Travel Time to Active Infection (+18.6%), Long-Range Intercity Gravity (+15.5%), Population Dynamics Embedding (PDFM (+14.8%), Local Economic Activity (GDP (-7.1%), Secondary Diffusion Pressure (+6.2%)
3	**19**	2	83	**Likati**	0.0340	0.0006	[0.0027 - 0.0749]	0.950	2	Travel Time to Active Infection (+21.4%), Flowminder Telecom Flow Pressure (+15.5%), Long-Range Intercity Gravity (+11.8%), Population Dynamics Embedd (PDFM) (+8.0%), Transit Hub Connectivi (+7.5%)
4	**23**	15	61	**Aketi**	0.0089	0.0008	[0.0027 - 0.0187]	0.902	10	Travel Time to Active Infection (+29.2%), Long-Range Intercity Gravity (+16.4%), Flowminder Telecom Flow Pressure (-7.8%), Secondary Diffusion Pressure (+7.3%), Population Dynamics Embeddi (PDFM) (-6.0%)
5	**25**	35	34	**Monga**	0.0044	0.0019	[0.0027 - 0.0091]	0.978	29	Travel Time to Active Infection (+23.5%), Long-Range Intercity Gravity (+12.8%), Flowminder Telecom Flow Pressure (-7.9%), Population Mobility Pressure (-7.6%), Socioeconomic Vulnerability Inc (+7.6%)
6	**33**	45	33	**Bondo**	0.0038	0.0019	[0.0027 - 0.0075]	0.898	35	Travel Time to Active Infection (+24.8%), Flowminder Telecom Flow Pressure (+14.1%), Secondary Diffusion Pressure (-6.4%), Distance to Outbreak Epicenter (-5.3%), Population Dynamics Embeddir (PDFM) (-5.2%)
7	**57**	42	108	**Bili (Bas-Uele)**	0.0040	0.0005	[0.0027 - 0.0073]	0.923	34	Travel Time to Active Infection (+25.4%), Secondary Diffusion Pressure (+9.4%), Socioeconomic Vulnerability Index (+9.3 Flowminder Telecom Flow Pressure (-7.9%), Long-Range Intercity Gravity (-6

** Top Contributing Features show signed percentage shares relative to total absolute feature attribution (100% × c_j / ∑|c_k|).*

Sud-Kivu Province

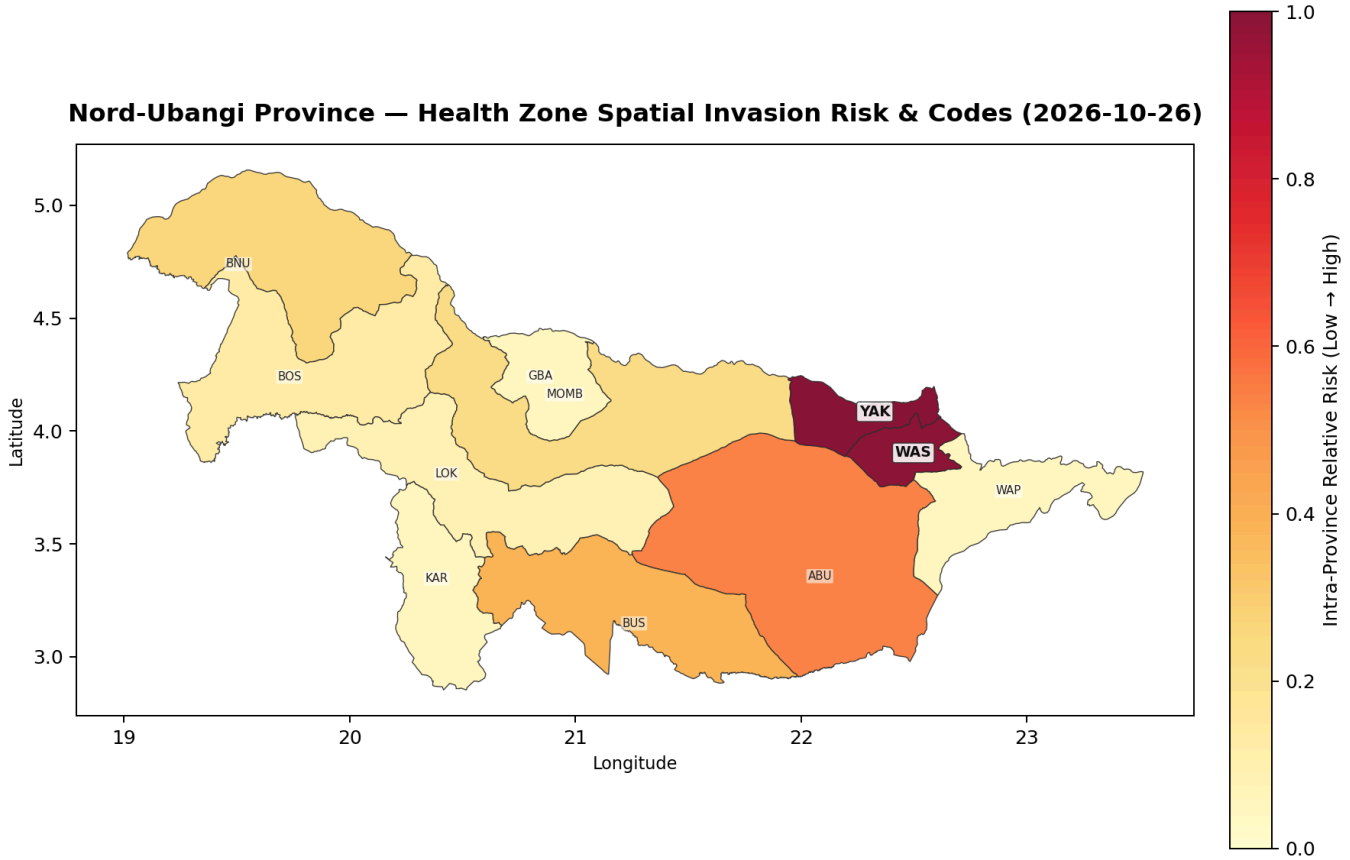


Top 10 Health Zones in Sud-Kivu (Ranked by RRF)

Prov #	RRF Rank	Google Rank	INRB Rank	Health Zone	Ensemble Score	INRB Pred	95% CI	Vulnerability (V)	Priority Rank	Top Contributing Features
1	**21**	19	47	**Kadutu**	0.0068	0.0014	[0.0027 - 0.0149]	0.218	30	Travel Time to Active Infection (+20.3%), Flowminder Telecom Flow Pressure (-12.3%), Distance to Nearest Active Zor (+8.4%), Secondary Diffusion Pressure (-6.9%), Long-Range Intercity Gravity (-5
2	**30**	18	69	**Minembwe**	0.0068	0.0008	[0.0027 - 0.0156]	0.693	19	Travel Time to Active Infection (+23.9%), Long-Range Intercity Gravity (+11.1%), Secondary Diffusion Pressure (+10.4%), Flowminder Telecom Flow Pressure (-7.8%), Population Mobility Pressure (-6.1%)
3	**32**	29	50	**Shabunda**	0.0053	0.0013	[0.0027 - 0.0104]	0.790	22	Flowminder Telecom Flow Pressure (+34.3%), Long-Range Intercity Gravity (-13.0%), Distance to Nearest Active Zor (-5.8%), Population Dynamics Embeddir (PDFM) (-5.5%), International Border Proximity (+4.9%)
4	**34**	80	17	**Uvira**	0.0027	0.0048	[0.0027 - 0.0046]	0.734	102	Travel Time to Active Infection (+21.3%), Long-Range Intercity Gravity (-11.5%), Flowminder Telecom Flow Pressure (-10.7%), Population Dynamics Embedd (PDFM) (-7.9%), Population Mobility Pressure (+5.7%)
5	**35**	68	22	**Ruzizi**	0.0028	0.0030	[0.0027 - 0.0059]	0.750	83	Travel Time to Active Infection (+23.6%), Long-Range Intercity Gravity (-12.8%), Population Dynamics Embedding (PDFM (+10.8%), Flowminder Telecom Flow Pressure (-8.9%), Mining Corridor Flow Hazard (-4.2%)
6	**47**	32	91	**Mulungu**	0.0048	0.0006	[0.0027 - 0.0088]	0.694	32	Travel Time to Active Infection (+28.6%), Flowminder Telecom Flow Pressure (-9.5%), Population Mobility Pressure (-7.1%), Urban Settlement Fraction (-6.6 Telecom Network Centrality (+6.5%)
7	**48**	106	31	**Ibanda**	0.0027	0.0020	[0.0027 - 0.0039]	0.308	387	Travel Time to Active Infection (+20.6%), Flowminder Telecom Flow Pressure (-10.4%), Long-Range Intercity Gravity (-10.1%), Secondary Diffusion Pressure (-8.7%), Distance to Nearest Active Zone (+6.1%)
8	**49**	26	131	**Haut Plateau**	0.0054	0.0004	[0.0027 - 0.0117]	0.699	25	Travel Time to Active Infection (+25.8%), Socioeconomic Vulnerability Index (+8.1 Flowminder Telecom Flow Pressure (-7.8%), Population Dynamics Embeddir (PDFM) (+6.7%), Long-Range Intercity Gravity (+5.5%)
9	**52**	84	49	**Nundu**	0.0027	0.0013	[0.0027 - 0.0042]	0.790	78	Travel Time to Active Infection (+26.7%), Secondary Diffusion Pressure (+8.9%), Flowminder Telecom Flow Pressure (-8.7%), Socioeconomic Vulnerability Inc (+7.4%), Population Mobility Pressure (-5.7%)
10	**60**	44	106	**Kimbi Lulenge**	0.0038	0.0005	[0.0027 - 0.0063]	0.734	38	Travel Time to Active Infection (+28.3%), Flowminder Telecom Flow Pressure (-9.4%), Population Mobility Pressure (-6.6%), Telecom Network Centrality (+5.7%), International Border Proximity (+5.6%)

** Top Contributing Features show signed percentage shares relative to total absolute feature attribution (100% × c_j / ∑|c_k|).*

Nord-Ubangi Province

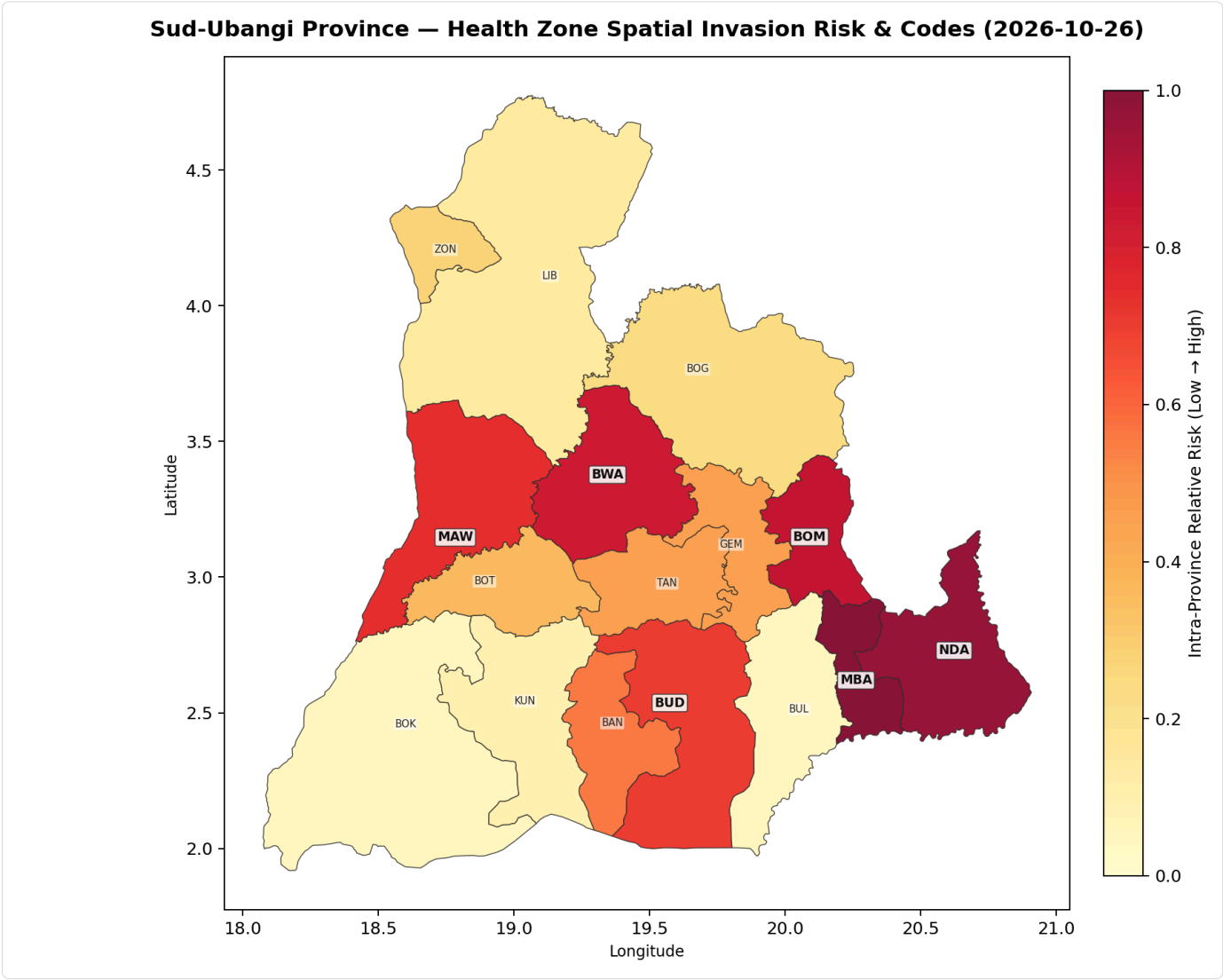


Top 10 Health Zones in Nord-Ubangi (Ranked by RRF)

Prov #	RRF Rank	Google Rank	INRB Rank	Health Zone	Ensemble Score	INRB Pred	95% CI	Vulnerability (V)	Priority Rank	Top Contributing Features
1	**3**	3	11	**Yakoma**	0.0275	0.0082	[0.0028 - 0.0552]	0.885	3	Flowminder Telecom Flow Pressure (+19.5%), Travel Time to Active Infectio (+19.2%), Transit Hub Connectivity (+8. Gravity Hazard (Power Law) (+5.8%), Socioeconomic Vulnerability Index (+5.
2	**4**	6	9	**Wasolo**	0.0134	0.0133	[0.0027 - 0.0257]	0.761	6	Travel Time to Active Infection (+25.5%) Long-Range Intercity Gravity (+14.6%), Socioeconomic Vulnerability Index (+8. Flowminder Telecom Flow Pressure (-6.8%), Gravity Hazard (Power Law) (+6.7%)
3	**36**	31	51	**Abuzi**	0.0048	0.0013	[0.0027 - 0.0104]	0.954	21	Travel Time to Active Infection (+20.1%) Secondary Diffusion Pressure (-17.8%), Population Dynamics Embedding (PDF (+12.0%), Long-Range Intercity Gravity (+6.7%), Flowminder Telecom Flow Pressure (-5.5%)
4	**50**	21	172	**Businga**	0.0063	0.0002	[0.0027 - 0.0118]	0.904	18	Travel Time to Active Infection (+27.7%) Population Dynamics Embedding (PDF (+14.0%), Long-Range Intercity Gravity (+9.7%), Flowminder Telecom Flow Pressure (-8.0%), Socioeconomic Vulnerability Index (+5.3%)
5	**85**	52	133	**Bili (Nord-Ubangi)**	0.0037	0.0004	[0.0027 - 0.0075]	0.943	36	Socioeconomic Vulnerability Index (+14.3%), Secondary Diffusion Pressure (+13.3%), Travel Time to Active Infectio (+10.7%), Flowminder Telecom Flow Pressure (-7.6%), Long-Range Intercity Gravity (+5.8%)
6	**90**	72	111	**Mobayi Mbongo**	0.0027	0.0005	[0.0027 - 0.0055]	0.728	106	Travel Time to Active Infection (+24.6%) Local Economic Activity (GDP pc) (-9.3% Flowminder Telecom Flow Pressure (-7.8%), Population Mobility Pressure (-7.2%), Source Epicenter Case Growth (+6.8%)
7	**128**	71	215	**Bosobolo**	0.0027	0.0002	[0.0027 - 0.0060]	0.953	58	Travel Time to Active Infection (+16.3%) Socioeconomic Vulnerability Index (+14.5%), Secondary Diffusion Pressure (-8.3%), Flowminder Telecom Flow Pressure (-7.5%), Spatial Gravity Hazard (-5.9%)
8	**137**	77	268	**Loko**	0.0027	0.0001	[0.0027 - 0.0052]	0.871	63	Travel Time to Active Infection (+24.9%) Flowminder Telecom Flow Pressure (-7.8%), Population Dynamics Embeddir (PDFM) (+7.5%), Population Mobility Pressure (-6.5%), Spatial Gravity Hazard (-5.8%)
9	**149**	124	177	**Karawa**	0.0027	0.0002	[0.0027 - 0.0030]	0.854	64	Travel Time to Active Infection (+24.5%) Long-Range Intercity Gravity (-8.0%), Flowminder Telecom Flow Pressure (-7.0%), Transit Hub Connectivity (+6.5% Spatial Gravity Hazard (-5.9%)
10	**152**	90	280	**Gbadolite**	0.0027	0.0001	[0.0027 - 0.0038]	0.691	119	Travel Time to Active Infection (+24.6%) Flowminder Telecom Flow Pressure (-8.6%), Spatial Gravity Hazard (-7.9%), Transit Hub Connectivity (+7.4%), Secondary Diffusion Pressure (-6.8%)

** Top Contributing Features show signed percentage shares relative to total absolute feature attribution (100% × c_j / ∑|c_k|).*

Sud-Ubangi Province



Top 10 Health Zones in Sud-Ubangi (Ranked by RRF)

Prov #	RRF Rank	Google Rank	INRB Rank	Health Zone	Ensemble Score	INRB Pred	95% CI	Vulnerability (V)	Priority Rank	Top Contributing Features
1	**89**	37	256	**Mbaya**	0.0044	0.0001	[0.0027 - 0.0074]	0.419	44	Travel Time to Active Infection (+27.9%), Secondary Diffusion Pressure (+16.4%), Long-Range Intercity Gravity (-6.8%), Flowminder Telecom Flow Pressure (-6.3%), Population Dynamics Embeddir (PDFM) (+5.1%)
2	**93**	38	278	**Ndage**	0.0043	0.0001	[0.0027 - 0.0086]	0.471	43	Travel Time to Active Infection (+27.4%), Secondary Diffusion Pressure (+10.6%), Flowminder Telecom Flow Pressure (-8.2%), Spatial Gravity Hazard (-5.9%), Local Economic Activity (GDP pc) (+5.7
3	**106**	53	229	**Bominenge**	0.0037	0.0002	[0.0027 - 0.0076]	0.434	59	Travel Time to Active Infection (+23.3%), Long-Range Intercity Gravity (+14.8%), Local Economic Activity (GDP pc) (-7.9 Source Epicenter Case Growth (+7.2%), Flowminder Telecom Flow Pressure (-6.
4	**107**	57	218	**Bwamanda**	0.0034	0.0002	[0.0027 - 0.0078]	0.622	54	Travel Time to Active Infection (+22.5%), Secondary Diffusion Pressure (+9.0%), Local Economic Activity (GDP pc) (-8.0 Source Epicenter Case Growth (+7.1%), Flowminder Telecom Flow Pressure (-6.
5	**122**	54	323	**Mawuya**	0.0036	0.0001	[0.0027 - 0.0057]	0.354	80	Travel Time to Active Infection (+25.2%), Secondary Diffusion Pressure (+14.5%), Long-Range Intercity Gravity (+8.4%), Flowminder Telecom Flow Pressure (-7.5%), Gravity Hazard (Power Law) (-5.
6	**129**	67	247	**Budjala**	0.0028	0.0001	[0.0027 - 0.0043]	0.573	159	Travel Time to Active Infection (+25.0%), Secondary Diffusion Pressure (-7.2%), Population Mobility Pressure (+6.8%), Spatial Gravity Hazard (-6.7%), Flowmin Telecom Flow Pressure (-6.2%)
7	**143**	76	332	**Bangabola**	0.0027	0.0001	[0.0027 - 0.0044]	0.521	255	Travel Time to Active Infection (+24.0%), Long-Range Intercity Gravity (+9.1%), Secondary Diffusion Pressure (-6.7%), Flowminder Telecom Flow Pressure (-6.4%), Spatial Gravity Hazard (-5.7%)
8	**161**	119	228	**Tandala**	0.0027	0.0002	[0.0027 - 0.0038]	0.544	239	Travel Time to Active Infection (+24.4%), Local Economic Activity (GDP pc) (-9.3 Flowminder Telecom Flow Pressure (-7.5%), Source Epicenter Case Growth (+7.0%), Secondary Diffusion Pressure (-4.8%)
9	**164**	209	128	**Gemena**	0.0027	0.0004	[0.0027 - 0.0027]	0.437	320	Travel Time to Active Infection (+16.7%), Population Dynamics Embedding (PDFM (-10.7%), Long-Range Intercity Gravity (-9.6%), Spatial Gravity Hazard (-6.6%), Flowminder Telecom Flow Pressure (-5.
10	**181**	115	327	**Boto**	0.0027	0.0001	[0.0027 - 0.0034]	0.479	285	Travel Time to Active Infection (+27.8%), Flowminder Telecom Flow Pressure (-9.1%), Gravity Hazard (Power Law) (-6.8%), Population Mobility Pressure (-6.5%), Spatial Gravity Hazard (-6.4%)

** Top Contributing Features show signed percentage shares relative to total absolute feature attribution (100% × c_j / ∑|c_k|).*

Ever-Infected Zones: 7-Day Case Outlook

Forecast Window: `2026-10-27` to `2026-11-02` | **Zones**: 66 ever-infected

A. Expected New Confirmed Cases, Active Zones

#	Health Zone	Province	Last 7d	Last 28d	Forecast 7d	80% Interval	Persistence
1	**Bunia ●**	Ituri	18	209	**36.2**	4–86	18
2	**Beni ●**	Nord-Kivu	15	209	**34.5**	2–85	15
3	**Mandima ●**	Ituri	33	249	**29.1**	2–72	33
4	**Katwa ●**	Nord-Kivu	4	92	**20.3**	1–49	4
5	**Rwampara ●**	Ituri	4	93	**17.9**	2–42	4
6	**Butembo ●**	Nord-Kivu	8	65	**13.6**	1–33	8
7	**Mabalako ●**	Nord-Kivu	7	54	**10.3**	0–25	7
8	**Nizi ●**	Ituri	4	61	**10.1**	1–24	4
9	**Wamba ●**	Haut-Uele	3	45	**9.3**	0–23	3
10	**Mambasa ●**	Ituri	3	48	**8.9**	0–22	3
11	**Kalunguta ●**	Nord-Kivu	5	30	**7.9**	0–19	5
12	**Nia Nia ●**	Ituri	4	42	**7.8**	0–20	4
13	**Oicha ●**	Nord-Kivu	2	30	**7.0**	0–18	2
14	**Musienene ●**	Nord-Kivu	0	31	**6.9**	0–17	0
15	**Komanda ●**	Ituri	0	23	**6.4**	0–16	0
16	**Lita ●**	Ituri	2	31	**6.4**	0–15	2
17	**Vuhovi ●**	Nord-Kivu	0	24	**5.7**	0–14	0
18	**Pawa ●**	Haut-Uele	1	24	**5.4**	0–13	1
19	**Mutwanga ●**	Nord-Kivu	4	21	**5.2**	0–13	4
20	**Mangala ●**	Ituri	0	32	**4.5**	0–11	0

● ≥ 10 cases in the last 28 days. Persistence = last 7 days repeated. 80% interval: 10th–90th percentile.

B. Hotspot Watch List (Ranked by P(≥5 New Cases))

Rank	Health Zone	Province	P(≥5)	P(≥1)	Last 7d
1	**Beni**	Nord-Kivu	**0.80**	1.00	15
2	**Bunia**	Ituri	**0.79**	1.00	18
3	**Mandima**	Ituri	**0.77**	0.99	33
4	**Katwa**	Nord-Kivu	**0.72**	0.99	4
5	**Rwampara**	Ituri	**0.69**	0.99	4
6	**Butembo**	Nord-Kivu	**0.64**	0.98	8
7	**Mabalako**	Nord-Kivu	**0.57**	0.97	7
8	**Nizi**	Ituri	**0.57**	0.98	4
9	**Mambasa**	Ituri	**0.54**	0.96	3
10	**Wamba**	Haut-Uele	**0.53**	0.95	3
11	**Kalunguta**	Nord-Kivu	**0.51**	0.96	5
12	**Nia Nia**	Ituri	**0.51**	0.95	4
13	**Oicha**	Nord-Kivu	**0.47**	0.94	2
14	**Komanda**	Ituri	**0.46**	0.95	0
15	**Musienene**	Nord-Kivu	**0.45**	0.92	0
16	**Lita**	Ituri	**0.44**	0.97	2
17	**Vuhovi**	Nord-Kivu	**0.40**	0.91	0
18	**Mutwanga**	Nord-Kivu	**0.39**	0.94	4
19	**Pawa**	Haut-Uele	**0.39**	0.91	1
20	**Mangala**	Ituri	**0.35**	0.93	0

C. Recurrence Watch: Ever-Infected Zones Without a Case Last Week

#	Health Zone	Province	P(≥1)	Last 28d
1	**Komanda**	Ituri	**0.95**	23
2	**Mangala**	Ituri	**0.93**	32
3	**Bambu**	Ituri	**0.93**	14
4	**Musienene**	Nord-Kivu	**0.92**	31
5	**Vuhovi**	Nord-Kivu	**0.91**	24
6	**Kyondo**	Nord-Kivu	**0.89**	11
7	**Nyankunde**	Ituri	**0.88**	6
8	**Gety**	Ituri	**0.87**	4
9	**Isiro**	Haut-Uele	**0.87**	6
10	**Rimba**	Ituri	**0.85**	7

** These lists cover zones that already reported a confirmed case. First invasions of never-infected zones are ranked in Section 1. Probabilities are model estimates, not calibrated frequencies. Models (best per task on backtest): A: Ensemble (production HGB, HGB + history, NB GLM); B ranking: Ensemble (production HGB, HGB + history, NB GLM); B P(≥1): Negative-Binomial GLM (own history + spatial); C: Negative-Binomial GLM (own history + spatial). Seeds: 20. Params: leaves=15, lr=0.01, iter=100.*

---

3. Limitations & Caveats

Spatial-Gravity Model Scope: The Google HistGBR model is optimized for contiguous geographic spread. It may underestimate risk for long-range stochastic jumps (e.g., via air travel or trader corridors) to distant, previously unaffected provinces.

Confirmed-Only Target: Invasion detection uses laboratory- confirmed cases only. Probable and suspected cases are used as input features but not as targets, which may delay detection of rapid invasion clusters.

Uncertainty Estimates: 95% CIs are empirical bootstrap intervals from 20 seeds. They reflect model variance but not epistemic uncertainty from reporting delays, missing data, or structural model limitations.

---

4. Google AI + INRB Reciprocal Rank Fusion (RRF k=60)

Multi-model ensemble combining Google HistGBR with published INRB SEIR model predictions.

RRF Rank	Health Zone	Province	Google Rank	INRB Rank	RRF Score
1	**Watsa**	Haut-Uele	7	2	0.031054
2	**Rethy**	Ituri	9	1	0.030886
3	**Yakoma**	Nord-Ubangi	3	11	0.029958
4	**Wasolo**	Nord-Ubangi	6	9	0.029644
5	**Bafwagbogbo**	Tshopo	1	16	0.029551
6	**Poko**	Bas-Uele	10	8	0.028992
7	**Makoro**	Haut-Uele	11	12	0.027973
8	**Biringi**	Ituri	20	6	0.027652
9	**Banalia**	Tshopo	13	13	0.027397
10	**Niangara**	Haut-Uele	27	4	0.027119
11	**Titule**	Bas-Uele	4	44	0.025240
12	**Pinga**	Nord-Kivu	5	43	0.025093
13	**Walikale**	Nord-Kivu	16	24	0.025063
14	**Doruma**	Haut-Uele	8	37	0.025015
15	**Aba**	Haut-Uele	30	15	0.024444
16	**Nyarambe**	Ituri	22	27	0.023689
17	**Kirotshe**	Nord-Kivu	17	35	0.023513
18	**Lubutu**	Maniema	25	28	0.023128
19	**Likati**	Bas-Uele	2	83	0.023122
20	**Opienge**	Tshopo	24	30	0.023016